LABORATORY REPORT: LAB 2 - LINEAR REGRESSION

Course: IT171IU · Statistical Learning

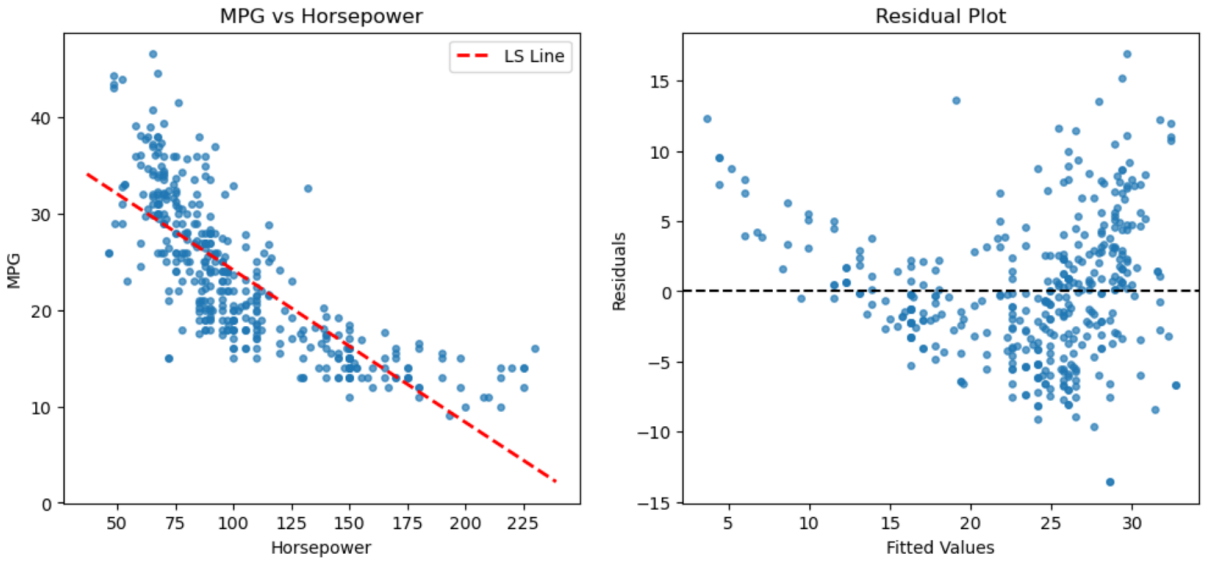
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Exercise 1 – Simple Regression on Auto

- **Model Fitted:** Simple linear regression of mpg on horsepower using statsmodels (sm.OLS).
- **Key Numbers & Coefficients:**
  - Intercept ($\hat{\beta}_0$): 39.9359 (Std Err: 0.717, t: 55.660)
  - Slope ($\hat{\beta}_1$): -0.1578 (Std Err: 0.006, t: -24.489)
  - $R^2 \approx 0.606$, Residual Standard Error (RSE) ≈ 4.90
- **Interpretation of Relationship:** There is a strong, statistically significant negative relationship between horsepower and fuel efficiency (p-value ≈ 0). As horsepower increases, miles per gallon decreases.
- **Predictions for horsepower = 98:**
  - Predicted Mean: ≈ 24.47 mpg
  - 95% Confidence Interval: $(24.03, 24.91)$ — quantifies the uncertainty of the *average* mpg for all cars with 98 HP.
  - 95% Prediction Interval: $(14.81, 34.13)$ — substantially wider because it accounts for individual observation noise (ε).
- **Diagnostics (Plots):** The scatter plot with the least-squares line shows a downward trend. However, the residual plot exhibits a clear U-shaped curved pattern,

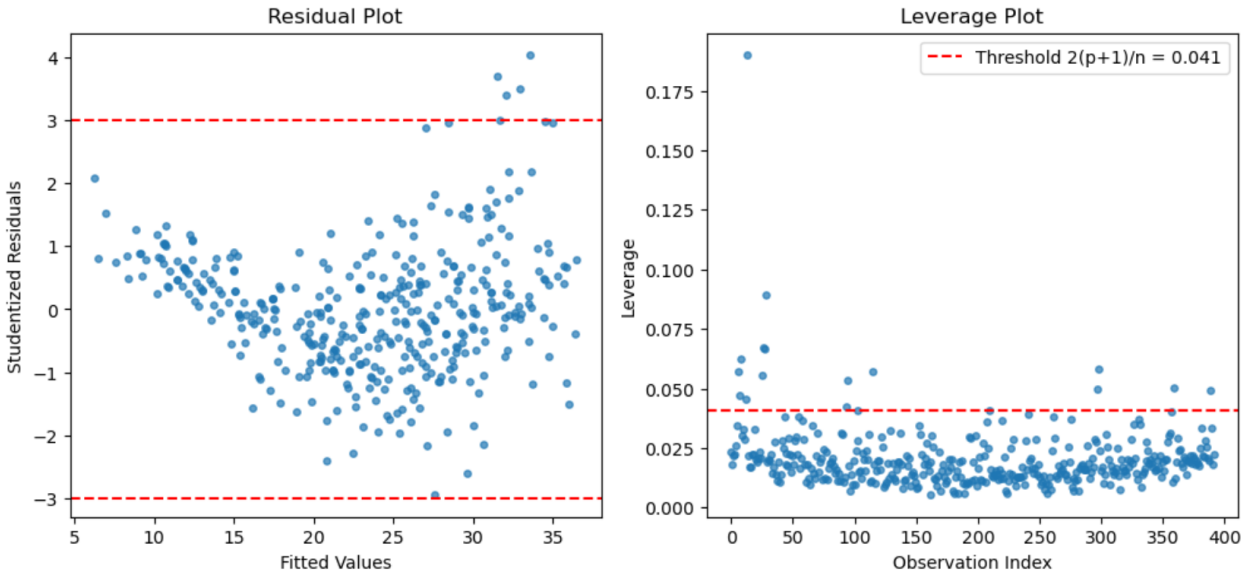
indicating non-linearity in the relationship.



Exercise 2 – Multiple Regression on Auto

- **Correlation Matrix Summary:** Quantitative variables like displacement, horsepower, and weight show very high positive correlations with each other and strong negative correlations with mpg.
- **Multiple Regression Findings:** When fitting mpg on all predictors except name, variables such as displacement, weight, year, and origin are statistically significant. The positive coefficient for year indicates that newer car models achieve better fuel efficiency over time, holding other factors constant.
- **Diagnostics (Residual & Leverage):**
    - Studentized residuals identify potential outliers exceeding ± 3.
    - Leverage values exceeding the threshold $2(p+1)/n$ point out observations with unusual predictor combinations that heavily influence the regression fit.
- **Collinearity (VIF):** Variables like displacement, horsepower, and weight yield Variance Inflation Factor (VIF) values well above 5 or 10, confirming strong

multicollinearity among engine and weight metrics.



Exercise 3 – Interactions and Transformations

- **Models Tested:**
    1. Base multiple regression model.
    2. Interaction model (adding ('horsepower', 'weight')).
    3. Polynomial model (adding poly('horsepower', degree=2)).
    4. Log-transformed horsepower model (log_horsepower).
- **Performance & ANOVA:** Adding polynomial terms or interaction effects increases R^2 compared to the base model. The anova_lm() comparisons yield significant p-values, proving that the expanded non-linear/interaction models provide a statistically significant improvement.
- **Selection Justification:** The transformed/interaction models are preferred because they successfully eliminate the non-linear curvature observed in the base residual plots.

Exercise 4 – Qualitative Predictors with Carseats

- **Model Fitted:** Sales regressed on Price, Urban, and US.
- **Coefficient Interpretations:**
    - Price: Negative coefficient; higher prices reduce unit sales.

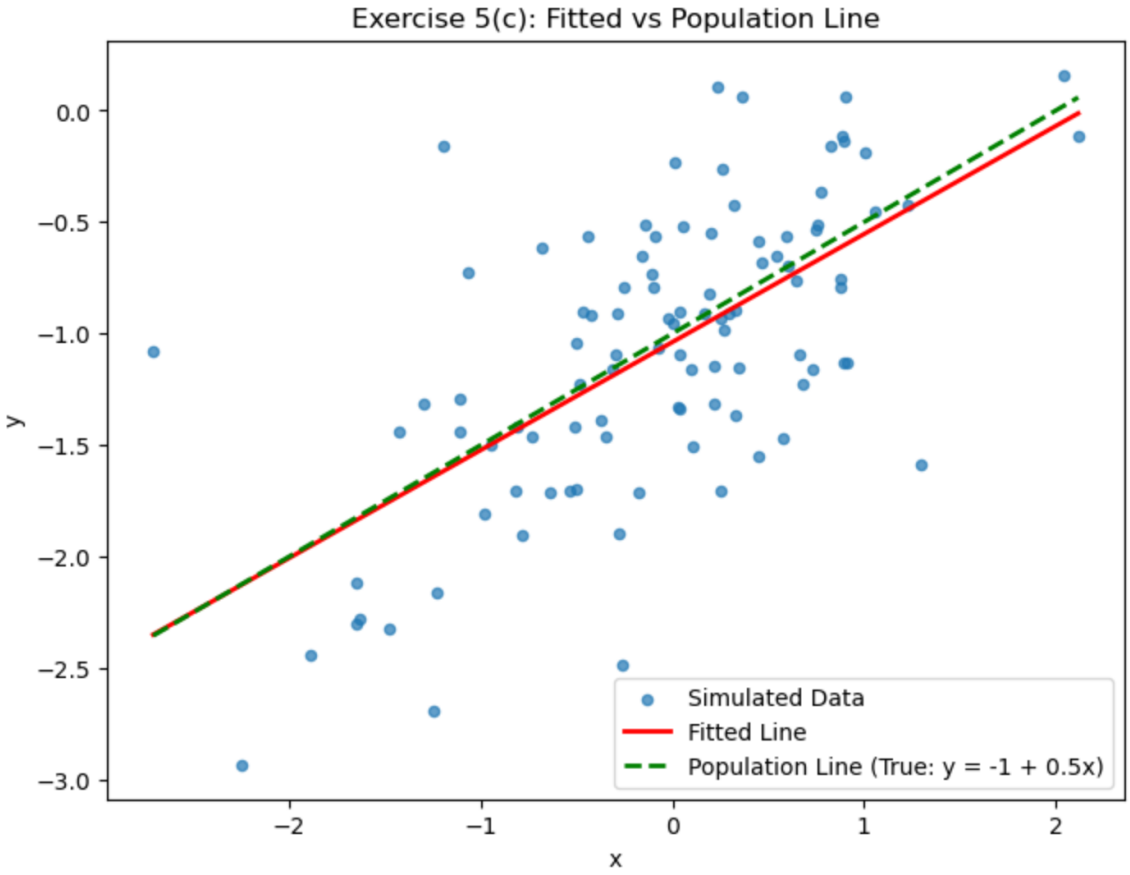
    - Urban: High p-value (not statistically significant), indicating no substantial difference in sales between urban and non-urban locations.
    - US: Positive coefficient; US store locations achieve higher sales on average compared to non-US locations.

- **Reduced Model & ANOVA:** Fitting a smaller model using only Price and US and comparing it with the full model via anova_lm() yields a large p-value, confirming that dropping the non-significant Urban predictor does not degrade model quality.
- **Confidence Intervals:** 95% confidence intervals for the smaller model confirm stable, tight bounds for Price and US.

Exercise 5 – Simulation Study

- **Simulation Setup:** Generated $n = 100$ samples with $x \sim N(0,1)$ and errors $\varepsilon \sim N(0,0.25)$, where $y = -1 + 0.5x + \varepsilon$.
- **Fit Results:** The estimated coefficients ($\hat{\beta}_0 \approx -1$, $\hat{\beta}_1 \approx 0.5$) closely recover the true population parameters.
- **Quadratic Term Test:** Adding a quadratic term x^2 yields a high p-value, correctly indicating that the quadratic term is unnecessary.

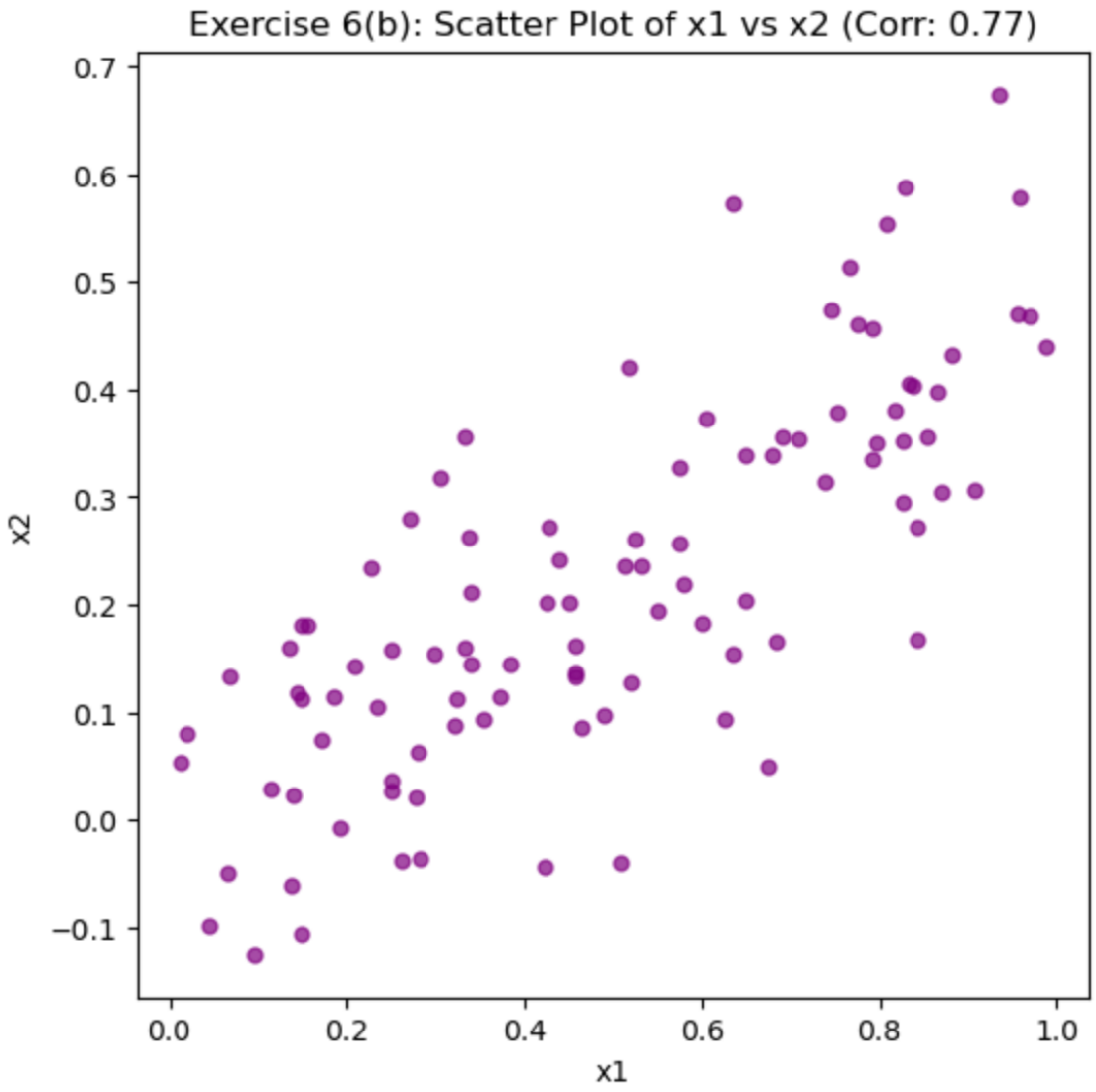
- **Noise Variance Effect:** Altering error variance ($\sigma^2 = 0.05$ vs 1.0) shows that higher noise widens the standard errors and confidence intervals of the slope coefficient.



Exercise 6 – Collinearity

- **Setup:** Generated y dependent on x_1 and a near-collinear predictor x_2 ($x_2 = 0.5x_1 + \text{noise}$).
- **Correlation:** The correlation between x_1 and x_2 is high.
- **Joint vs. Individual Fits:** When included together in the multiple regression model, collinearity inflates standard errors, often causing one or both variables to lose statistical significance despite being highly significant when fitted individually. High

VIF values confirm this issue.



Exercise 7 – Additional Data Set: Diabetes Progression

- **Models Compared:**
  - Model 1 (bmi alone): Lower R^2, higher RSE.
  - Model 2 (bmi, bp, s5): Moderate R^2 improvement.
  - Model 3 (All 10 predictors): Highest R^2 and lowest RSE.
- **ANOVA Test:** Comparing Model 2 and the full model via anova_lm() yields a small p-value, showing that the full model significantly improves explanatory power.

- **Collinearity (VIF):** Several blood serum measurements (s1–s6) exhibit high VIF values, indicating multicollinearity among lab assays.
- **Clinical Report Recommendation:** Predictors such as bmi, s5, and bp (or selected serum metrics) are most critical to report to a medical professional based on their magnitude and statistical significance.